Cross-Country Term Life Insurance Pricing

A Mortality-Driven Premium Analysis: India, Japan, and the United States

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1. Overview

This project builds a term life insurance pricing model for a standard policy across three countries - India, Japan, and the United States using publicly available mortality data. The central question is straightforward: if the same policy is sold to a 30-year-old male in each country, how much should the insurer charge and what drives the difference?

The answer lies almost entirely in mortality. By holding every other variable constant like sum assured, policy term, discount rate, entry age the model isolates the effect of mortality on premium pricing. The results are significant: Japan's premium works out to roughly a third of the equivalent US figure, with India sitting in between.

2. Policy Parameters

The following assumptions apply uniformly across all three countries:

Parameter	Assumption
Policy Type	20-Year Level Term
Sum Assured	₹50,00,000
Entry Age	30 years (male)
Discount Rate	6% per annum
Mortality Basis	Ultimate table

Using a single discount rate and entry age across all three countries keeps the comparison clean. Any premium difference is attributable to mortality alone, which is exactly what the model is designed to measure.

3. Data Sources and Methodology

Mortality Tables

Each country uses a different data source, reflecting what is available and what practitioners actually use:

- United States: Human Mortality Database (HMD), period life table, males, 2019. The HMD is maintained by UC Berkeley and the Max Planck Institute and is the international standard for demographic mortality research.

- Japan: Human Mortality Database (HMD), period life table, males, 2019. Japan consistently records some of the lowest mortality rates globally, particularly at working ages.
- India: Institute of Actuaries of India (IALM) 2006-08 ultimate table. This is the most recent published assured lives table for India and remains the standard basis used by IRDAI-regulated insurers. No equivalent HMD data exists for India.

2019 data was selected for USA and Japan to avoid the distortion in mortality rates caused by the COVID-19 pandemic in 2020 and 2021. This is consistent with standard actuarial practice when constructing pre-pandemic pricing bases.

Pricing Methodology

Premiums are calculated using the equivalence principle, which is the standard net premium basis in actuarial science. Under this principle, the expected present value of future premiums equals the expected present value of future claims at inception.

Net Annual Premium = EPV of Claims / Annuity Factor

Where the EPV of claims sums discounted expected death benefits over the 20-year term, and the annuity factor is the sum of discount factors over the same period. Both are calculated from the relevant life table using a 6% discount rate.

4. Results

Net Annual Premiums

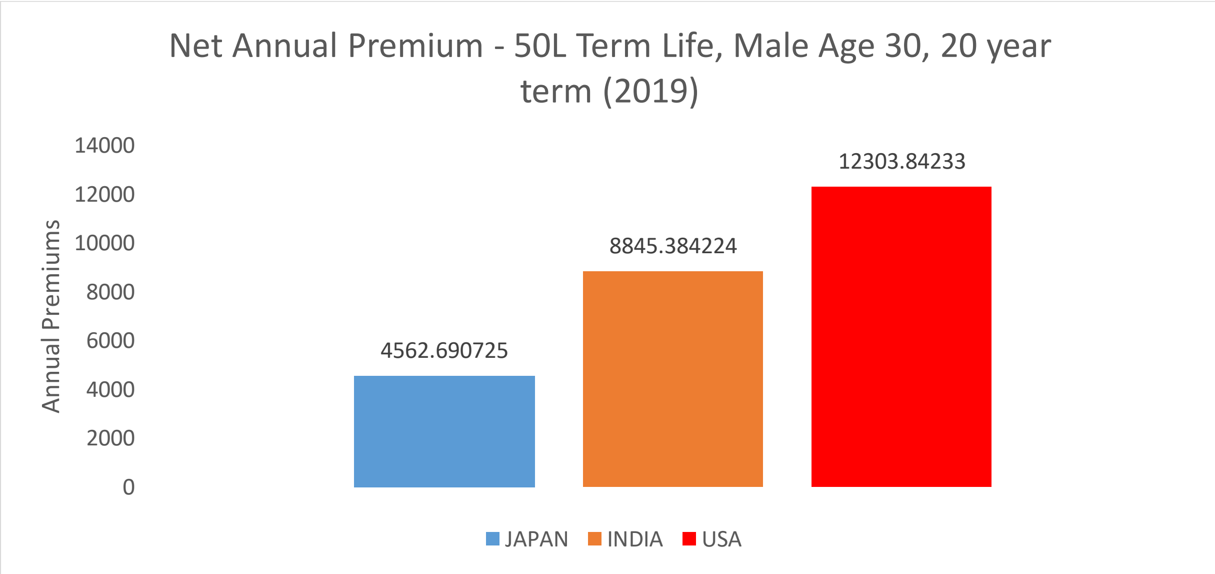
Country	Data Source	Net Annual Premium	vs Japan
Japan	HMD 2019	₹4,563	Baseline
India	IALM 2006-08	₹8,845	+94%
USA	HMD 2019	₹12,304	+170%

The figures above are net premiums only. They do not include expense loadings, profit margins, or reinsurance costs, which would be added in a commercial pricing exercise. The purpose here is to isolate the pure mortality component of the premium.

Key Observations

- The USA premium is 2.7 times the Japanese equivalent for an identical policy. This is driven by significantly higher mortality rates at every age in the 30-49 range a pattern consistent with higher rates of cardiovascular disease, obesity, and accidental death in the US population.

- India sits between the two, which is expected. The IALM table reflects assured lives - people who have been medically underwritten and therefore shows lower mortality than the general Indian population. This makes the India-Japan comparison closer than it would be on a population basis.
- Japan's lx at age 49 (end of policy term) is 97,186, compared to 92,543 for the USA. This means roughly 5% more of the Japanese cohort survives to the end of the policy term, a substantial difference over 20 years.



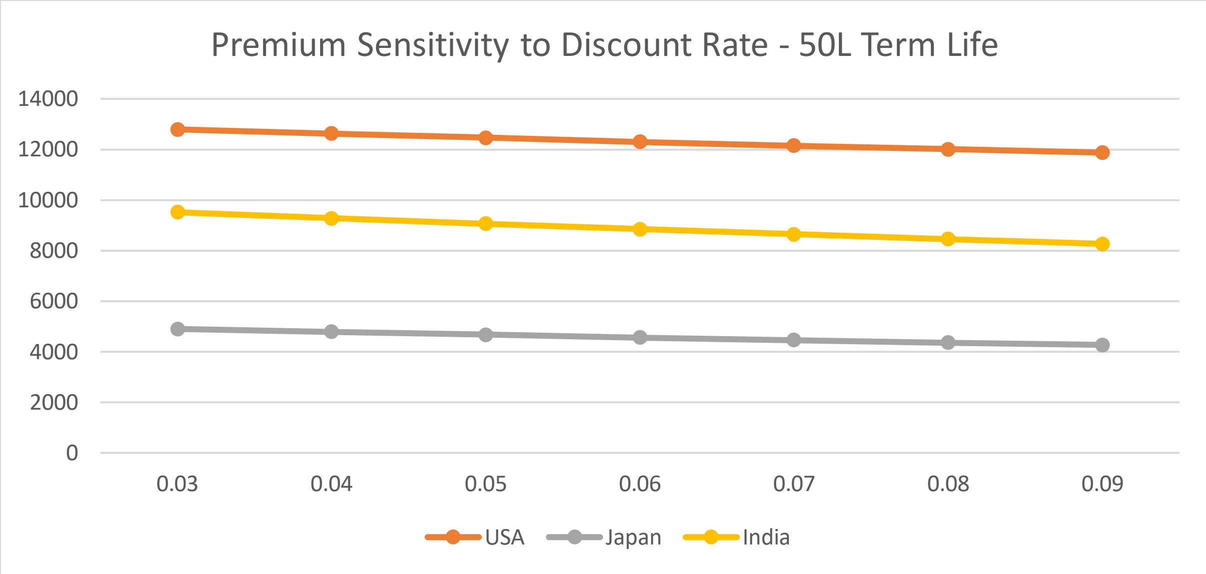
5. Sensitivity Analysis

Interest Rate Sensitivity

Premiums were recalculated across a range of discount rates from 3% to 9% to assess how sensitive each country's premium is to interest rate assumptions.

Discount Rate	USA	Japan	India
3%	₹12,794	₹4,900	₹9,512
4%	₹12,624	₹4,783	₹9,280
5%	₹12,461	₹4,670	₹9,058
6% (base)	₹12,304	₹4,563	₹8,845
7%	₹12,154	₹4,460	₹8,643
8%	₹12,011	₹4,363	₹8,451
9%	₹11,874	₹4,271	₹8,268

Across the full rate range tested, India shows the largest absolute premium change at ₹1,244, compared to ₹920 for the USA and ₹629 for Japan. As a proportion of the base premium, the sensitivity is broadly similar across countries. Premiums fall as rates rise because higher investment returns reduce the present value of future claims.

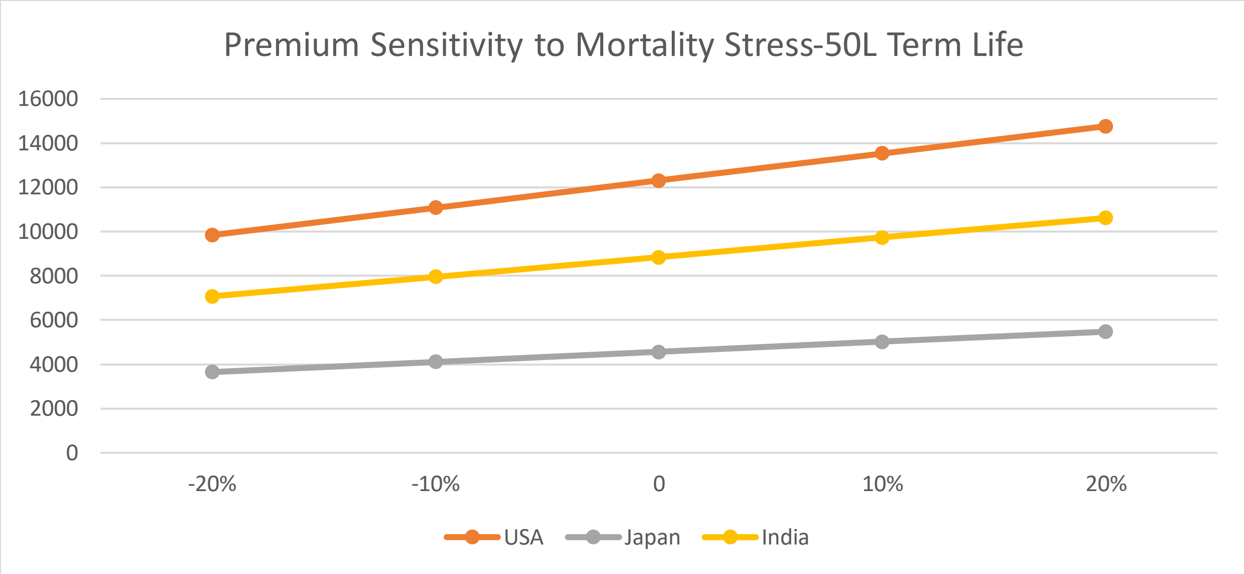


Mortality Stress Testing

A parallel stress test applies uniform percentage shocks to the underlying mortality rates to simulate scenarios such as pandemic excess mortality or long-run improvements in longevity.

Mortality Stress	USA	Japan	India
-20% (improvement)	₹9,843	₹3,650	₹7,076
-10%	₹11,073	₹4,106	₹7,961
0% (base)	₹12,304	₹4,563	₹8,845
+10%	₹13,534	₹5,019	₹9,730
+20% (deterioration)	₹14,765	₹5,475	₹10,614

The USA shows the largest absolute swing across the stress range at ₹4,922, reflecting its higher base mortality. In proportional terms, however, all three countries respond almost identically to the same mortality shock confirming that the premium differences between countries are structural, driven by the level of mortality rather than its volatility.



6. Limitations

- The IALM 2006-08 table is the most recent available for India, but is now approximately 17 years old. Mortality in India has likely improved since then, which would reduce the India premium if more current data were available.
- The model uses general population data for the USA and Japan but assured lives data for India. This creates a like-for-like inconsistency that overstates the India-Japan gap relative to a population-level comparison.
- No mortality improvement factors have been applied. In a commercial pricing exercise, projected improvements in longevity would typically be incorporated using scales such as the CMI model (UK) or SOA MP scales (USA).
- Premiums are net of expenses. Office premiums - which include loadings for administration, commission, and profit that would be materially higher and vary by distribution channel and market.
- The model assumes all policyholders pay premiums throughout the term and does not account for lapses, which in practice reduce the insurer's risk exposure at later durations.

7. Potential Extensions

- Apply mortality improvement factors to project future mortality and assess the impact on long-duration pricing.
- Calculate office premiums by adding expense loadings and a profit margin, bringing the model closer to a commercially usable basis.
- Extend the analysis to whole life or endowment products, where the sensitivity to both mortality and interest rate assumptions is significantly greater.
- Incorporate select mortality to model the effect of underwriting, reflecting the lower initial mortality of newly assured lives versus the ultimate rates used here.

Data sources: Human Mortality Database (mortality.org); Institute of Actuaries of India, IALM Mortality Tables 2006-08. All calculations performed in Microsoft Excel using the equivalence principle on a net premium basis.